THE DELTA-INVARIANT OF A REDUCED ONE-DIMENSIONAL NAGATA LOCAL RING IS UNCHANGED BY HENSELIZATION, STRICT HENSELIZATION AND COMPLETION

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Allowed standard prerequisites (editorial).

- Normalization commutes with completion. Stacks Project, Varieties, Lemma lemma-normalization-same-after-completion (tag 0C3V). Identifies the integral closure of the completion with the base change of the integral closure.
- Length of a base-changed quotient module. Stacks Project, Algebra, Lemma algebra-lemma-pullback-module (tag 00P7). Computes the length of B'/B from the length of A'/A.

Source reference. Original lemma 0C3V: sources/stacks-chow/tags.txt, source lines 15205–15205. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A be a reduced Nagata local ring of dimension 1. The δ-invariant of A is the same as the δ-invariant of the henselization, strict henselization, or the completion of A.*

Proof. Let us do this in case of the completion $B = A^\wedge$; the other cases are proved in exactly the same manner. Let A', resp. B' be the integral closure of A, resp. B in its total ring of fractions. Then $B' = A' \otimes_A B$ by Lemma 0C3V. Hence $B'/B = (A'/A) \otimes_A B$. The equality now follows from Algebra, Lemma 02M1 and the fact that $B \otimes_A \kappa_A = \kappa_B$. □

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